Select ten contacts from 1,000?

Same ten lowest-Neff eval-val proteins and two MarinFold checkpoints.

20,000 randomized contact rollouts; 18,000 newly generated on CoreWeave H100s.

Fold ten contacts per arm: first ten, random ten, cluster medoids, farthest points.

600 new Modal Helico predictions; 800 Helico plus 300 earlier baseline structures.

Selection inputs were frozen and published before their folding calls.

Medoids cover the sampled pool better, but reduce pairwise contact separation.

A secondary farthest-point arm explicitly raises separation from ~0.94 to ~0.98.

Matched Helico seeds, six recycles, one sample, unchanged checkpoint and scorer.

Medoids: promising default oracle

Default random -> medoids: oracle TM 0.462 -> 0.528; confidence 0.407 -> 0.469.

Oracle gain +0.066 [0.020, 0.121]; confidence gain +0.062 [-0.017, 0.160].

MPNN random -> medoids: oracle 0.500 -> 0.488; confidence 0.440 -> 0.434.

MPNN confidence gain -0.007 [-0.096, 0.088]. No established primary gain.

Default medoids also exceed first-ten confidence 0.433, but that interval spans zero.

8eb9_A drives much of the gain: default random 0.315 versus medoids 0.743.

All intervals: 10,000 paired protein-bootstrap resamples, conditional on one control draw.

Oracle is best of ten folded selections, not best of 1,000 structures.

More separation hurts typical folds

Default random -> farthest: mean TM 0.336 -> 0.273; median 0.326 -> 0.252.

MPNN random -> farthest: mean TM 0.343 -> 0.264; median 0.335 -> 0.243.

Mean TM changes: -0.063 [-0.087, -0.039] and -0.079 [-0.118, -0.041].

Mean/median lDDT and GDT-TS also decline for both checkpoints.

Confidence TM changes -0.016 / -0.013; both intervals include zero.

No observed oracle or confidence-pick TM gain from farthest-point selection.

This tests two specific selectors; contact counts also change. Ten-protein pilot.

Use coverage-oriented selection as a hypothesis to test further, not a proven gain.

Diversity experiment: evidence and cost

All 600 new structures scored; full native CA coverage; >=99.584% heavy atoms.

One pinned Helico runtime and checkpoint, same structural scorer for all arms.

New contact inference: 538.0 H100-seconds; Helico inference: 5,041.1 seconds.

Model setup/staging and container overhead are additional; timings are not billing.

Medoid clustering: 2.50 CPU-seconds. Fourteen Helico workers including checks.

Public HF v3-diversity preserves raw pools, chosen contacts, structures and hashes.

See README, per-protein CSVs, paired contrasts and three new figures.

Earlier ESMFold2 / AF3 baselines follow. They were not rerun for this extension.

Five arms, 500 structures

Ten lowest-Neff natural eval-val proteins; ten independent samples per arm.

Default MarinFold and MPNN-only two-epoch lr5e-4 each supply contacts to Helico.

Added 100 fresh predictions each: ESMFold2, AF3 single sequence, AF3 with MSA.

All arms use the same resolved sequences, native references and scorer.

All 500 predictions are complete; every structure has full native CA coverage.

AF3 + MSA leads on this pilot

Confidence-selected macro means: TM-score / all-heavy-atom lDDT / GDT-TS.

AF3 + MSA: 0.799 / 0.730 / 0.780.

ESMFold2: 0.640 / 0.565 / 0.605.

MPNN-only MarinFold -> Helico: 0.440 / 0.397 / 0.373.

Default MarinFold -> Helico: 0.433 / 0.388 / 0.378.

AF3 single sequence: 0.403 / 0.370 / 0.358.

Compared with default Helico, paired TM-score differences (95% protein bootstrap):

ESMFold2 +0.207 [0.034, 0.370]; AF3 + MSA +0.366 [0.169, 0.541].

AF3 single sequence -0.030 [-0.099, 0.044]; no clear difference in ten proteins.

Sampling helps Helico, but leaves a gap

Oracle / median / confidence / individual-mean TM-score at ten samples:

Default -> Helico: 0.483 / 0.340 / 0.433 / 0.347.

MPNN-only -> Helico: 0.481 / 0.350 / 0.440 / 0.348.

ESMFold2: 0.672 / 0.625 / 0.640 / 0.626.

AF3 single sequence: 0.465 / 0.413 / 0.403 / 0.413.

AF3 + MSA: 0.838 / 0.801 / 0.799 / 0.793.

Helico's oracle-over-mean gain is about 0.135, versus 0.045-0.052 for baselines.

Yet ESMFold2's individual mean exceeds Helico's observed oracle best-of-ten.

The two MarinFold checkpoints still have no clear aggregate winner.

Confidence helps some models more

TM-score gain over the individual-sample mean, with 95% protein intervals:

Default -> Helico: +0.086 [0.017, 0.159].

MPNN-only -> Helico: +0.092 [0.037, 0.154].

ESMFold2: +0.015 [0.002, 0.031].

AF3 single sequence: -0.010 [-0.032, 0.010].

AF3 + MSA: +0.006 [-0.016, 0.034].

This pilot does not demonstrate a confidence-selection gain for either AF3 arm.

ESMFold2 uses pTM; AF3 and Helico use their own native ranking scores.

Each confidence choice supplies all metrics; oracle is separately best per metric.

The lowest-Neff target is an exception

8eb9_A, Neff 1.34: confidence-selected TM-score is 0.728 for MPNN -> Helico.

Default Helico 0.531; AF3 single sequence 0.519; ESMFold2 0.255; AF3 + MSA 0.233.

8bux_A: all single-sequence arms score 0.157-0.218, but AF3 + MSA scores 0.922.

Low Neff does not mean that the alignment is useless, or that it always helps.

AF3 MSA-minus-single-sequence mean TM-score: +0.396 [0.196, 0.565].

Ten proteins cannot establish full eval-val performance or a universal ranking.

Matched inputs and structural scoring

Baseline seeds: 381000 + 100*cohort_index + sample_index; one sample per seed.

AF3: official pinned source, ten recycles, templates off in both arms.

MSA: archived ColabFold rows projected onto the same resolved residues.

ESMFold2: released pinned snapshot, 20 loops, 200 diffusion steps, LM dropout .3.

TMscore: reference-normalized TM-score and optimized GDT-TS, fixed residue IDs.

lDDT: all native heavy atoms, 15 A neighborhoods, within-residue pairs excluded.

Missing atoms receive identical penalties; minimum baseline coverage is 99.57%.

This matches sample budgets, not compute or model training exposure.

Reproduction and compute

README, per-sample CSVs, all prefix summaries and plotting scripts are in exp381.

Public HF v1 retains the original 200 Helico predictions and native references.

Public v2-baselines adds all 300 baseline CIFs, mapped arrays and input alignments.

Run analyze_baselines.py and build_summary.py for figures/PDF without GPU access.

Measured new inference: 1,407.6 H100-seconds; shared setup: 240.3 seconds.

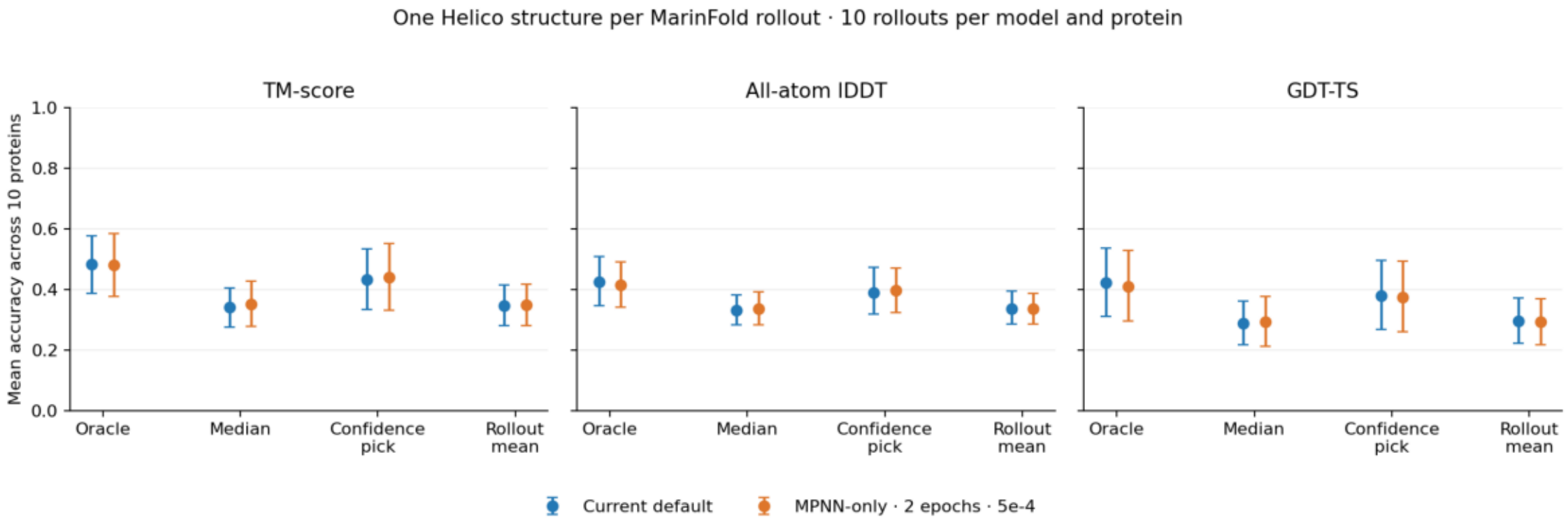
AF3 includes cold-shape JIT; ESM folding includes preparation. Not a speed test.

Eleven workers including smoke; all full-budget prediction calls succeeded.

See README for pinned revisions, Modal runs, source hashes and limitations.

accuracy_summary.png

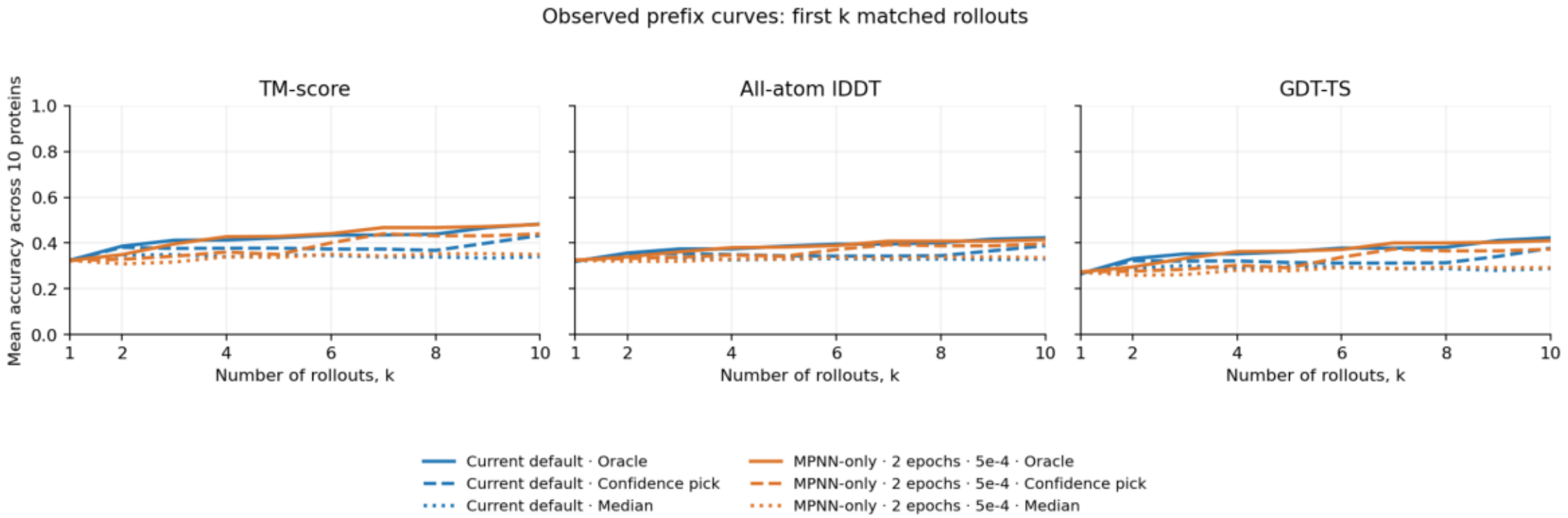
Equal-protein means at k=10; 95% protein-bootstrap intervals (10,000 draws). Oracle uses reference labels; confidence pick uses only Helico ranking_score.



$ analyze.py

accuracy_vs_rollouts.png

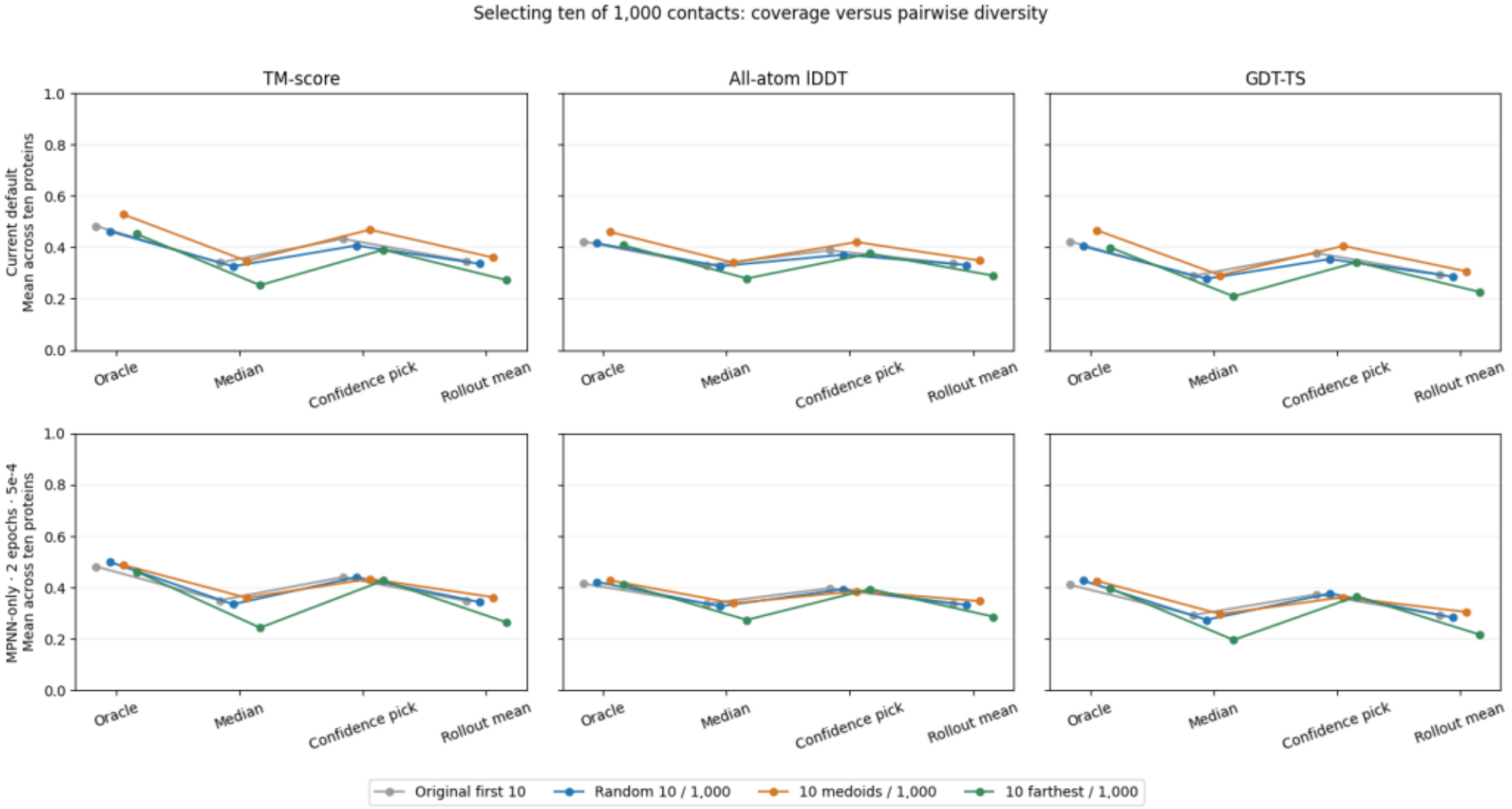
Oracle, confidence-selected, and median accuracy for the first k saved rollouts. Lines describe one fixed sequence of ten samples, not expected curves over all possible rollout orders.




```
$ analyze.py
```

diversity_accuracy.png

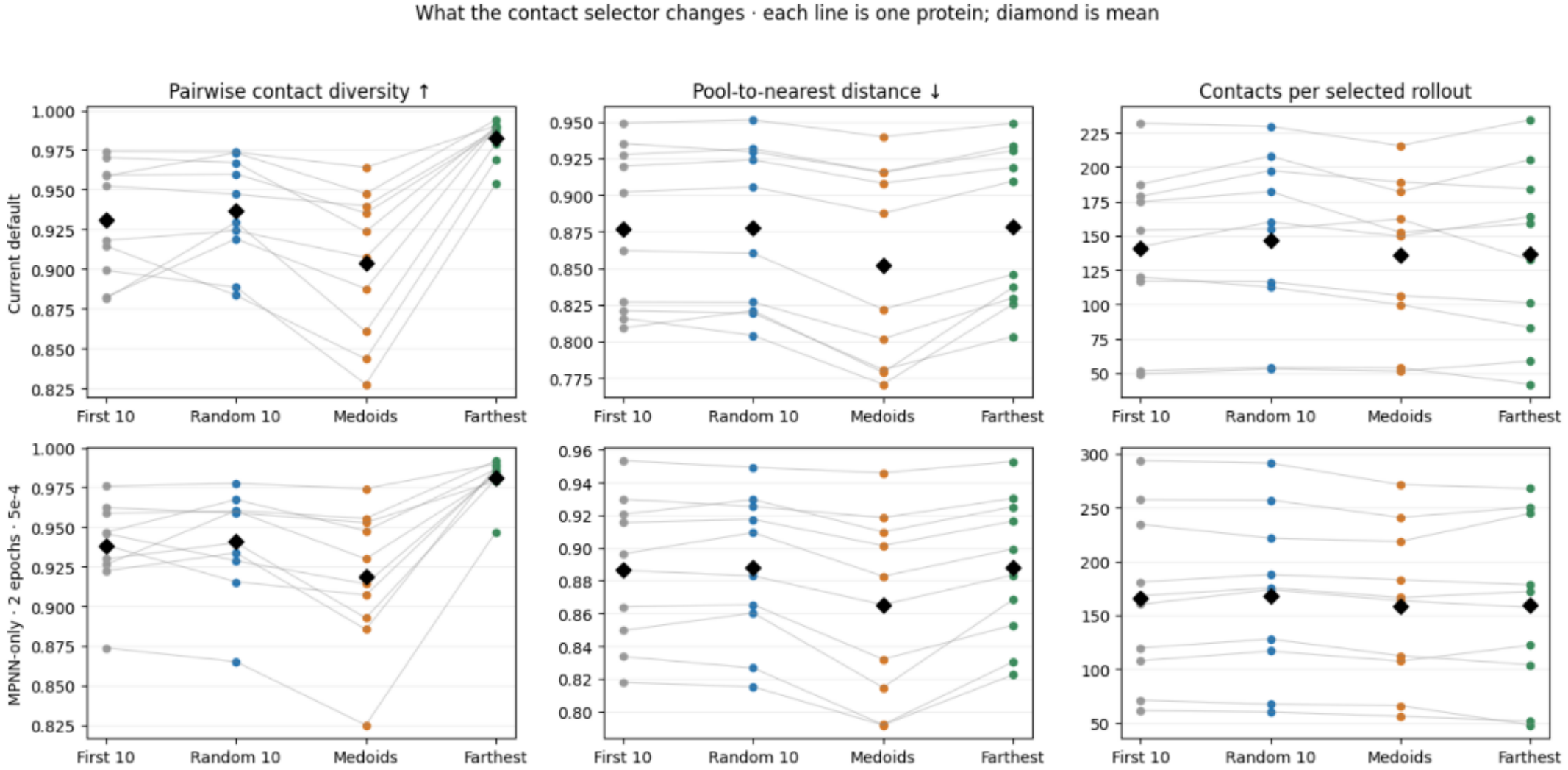
All arms fold ten contact sets per protein with matched per-slot Helico seeds. Mean across the same ten lowest-Neff eval-val proteins. Medoids, farthest points and random controls use a 1,000-rollout pool per checkpoint; original first ten are reused.




```
$ analyze_diversity.py
```

diversity_contact_geometry.png

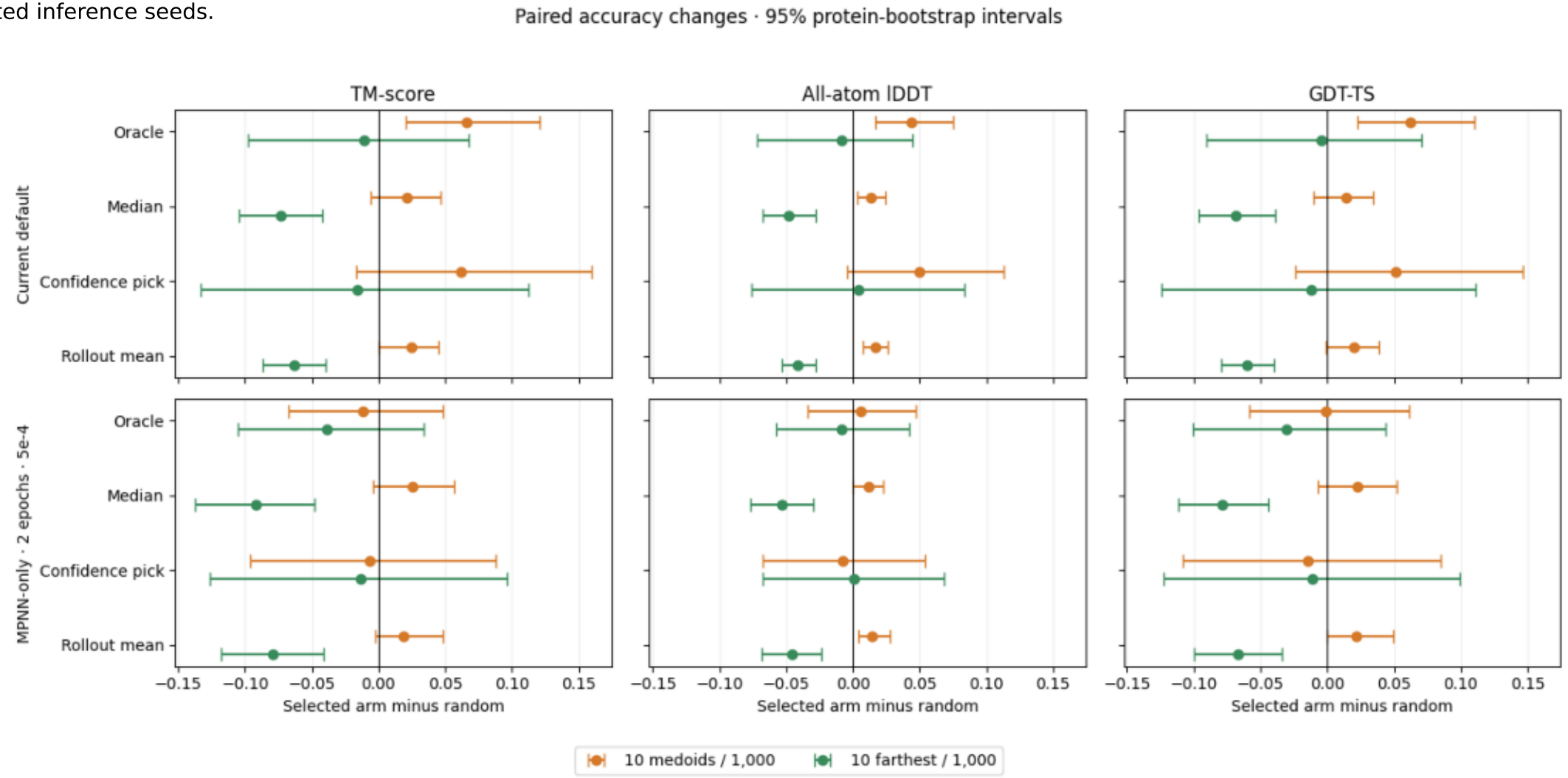
Jaccard distance on the exact contact sets supplied to Helico. Medoids optimize mean nearest-representative distance, rather than directly maximizing pairwise separation. All 1,000 draws retain their sampling weight, including duplicates.




```
$ analyze_diversity.py
```

diversity_paired_changes.png

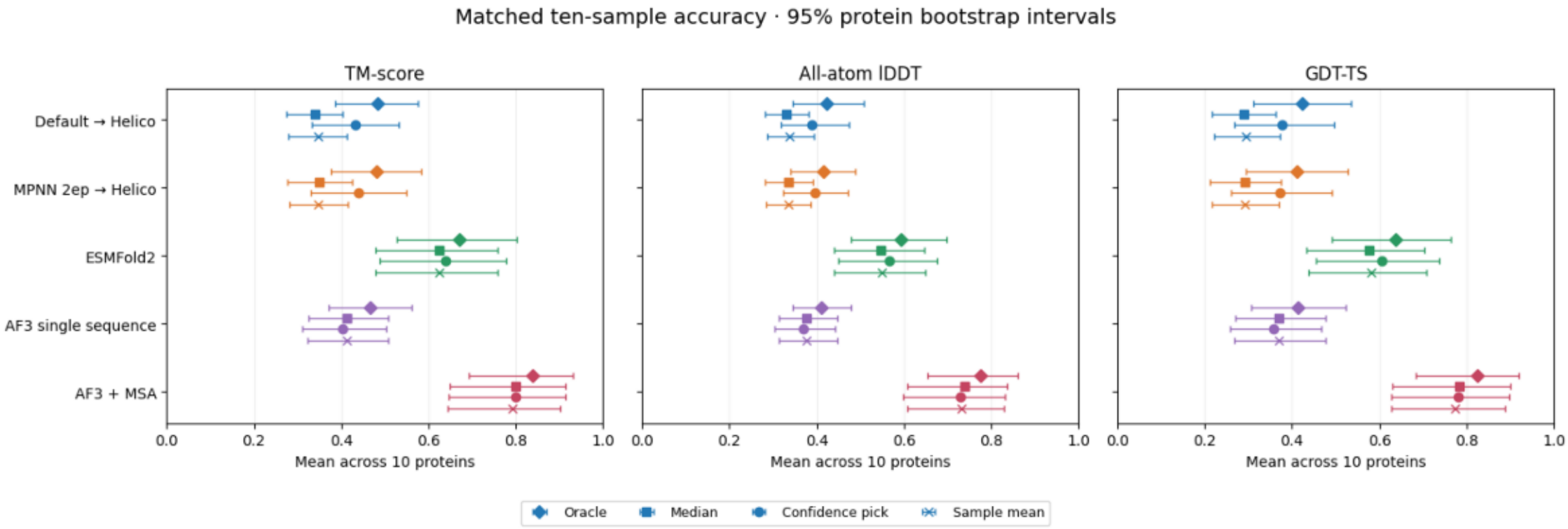
10,000 paired bootstrap resamples of ten proteins (seed381). Medoid confidence-selected TM-score is the original primary endpoint; farthest-point comparisons are secondary. Intervals describe protein variation conditional on one fixed random subset and the selected inference seeds.




```
$ analyze_diversity.py
```

matched_accuracy_summary.png

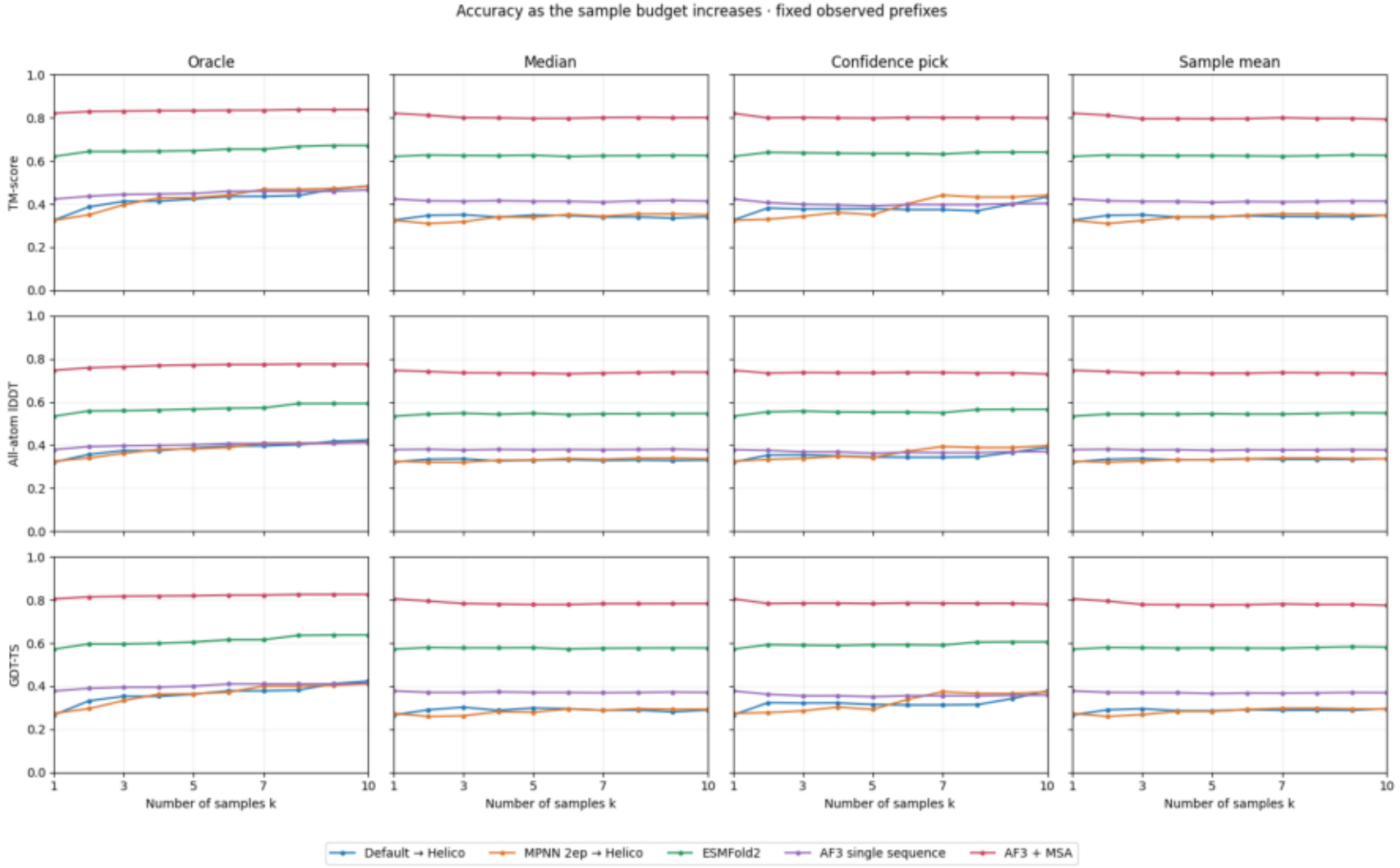
All five arms use the same ten proteins and ten samples each. Oracle is metric-specific; confidence selects one structure.
Whiskers bootstrap proteins, not rollouts.



$ analyze_baselines.py

matched_accuracy_vs_samples.png

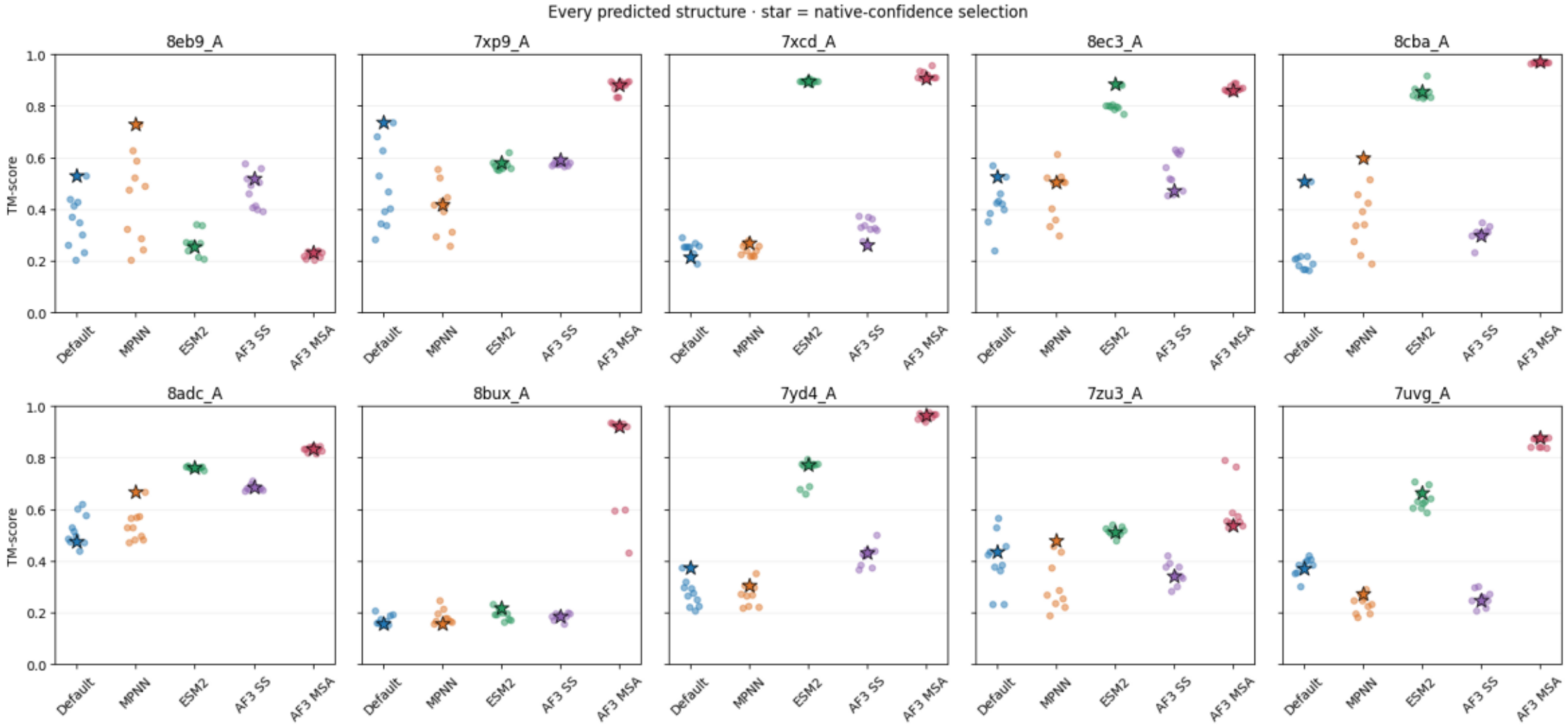
Macro averages of all fixed prefixes k=1..10. Only oracle is monotonic; confidence pick may worsen as more samples arrive. Sample mean is the per-protein individual-sample average.



$ analyze_baselines.py

matched_per_protein_tm.png

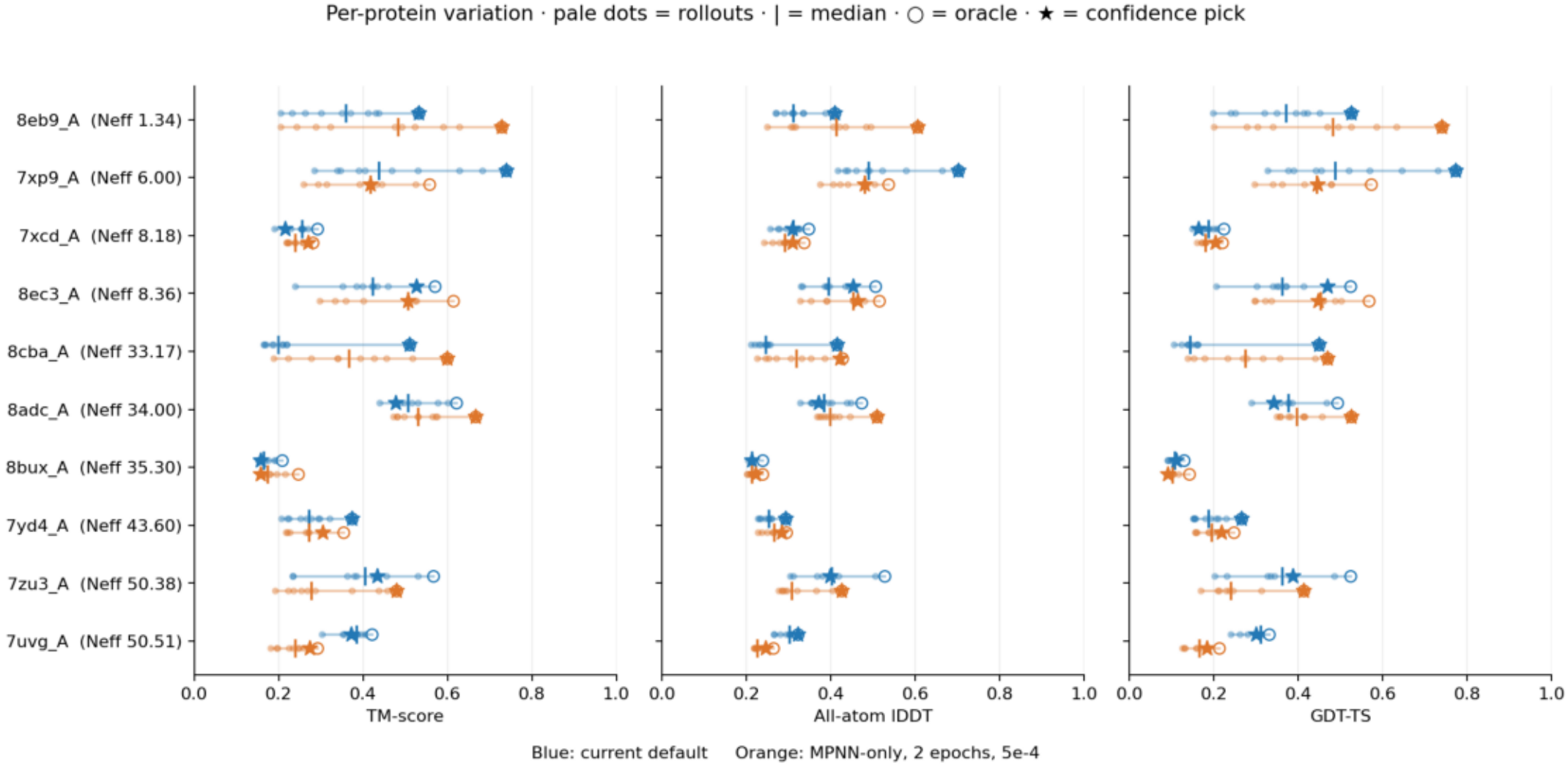
Ten samples per protein per arm; star identifies the single confidence-selected structure. Targets appear in increasing original Neff order.




```
$ analyze_baselines.py
```

per_protein_accuracy.png

Every individual prediction, with median, oracle, and confidence-selected accuracy. Targets are ordered by Neff. Each structure contributes the same three metrics.




```
$ analyze.py
```